

MUST lecture 7

Multivariate Multiple Linear Regression

(MMLR)

aka. GLM, General Linear Model

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

MMLR, Multivariate Multiple Linear Regression aka. GLM, General Linear Model

UMLR model

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & z_{11} & \dots & z_{1r} \\ 1 & z_{21} & \dots & z_{2r} \\ \vdots & \vdots & & \vdots \\ 1 & z_{n1} & \dots & z_{nr} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_r \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

Y = univariate response variable (known/measured)

$z_1, \dots, z_r$ = predictor variables with known (set) values

$\beta_0, \beta_1, \dots, \beta_r$ = unknown regression coefficients (marginal slopes)

ε = random error term for non-perfect model (unknown)

MMLR model

$$\begin{bmatrix} Y_{11} & Y_{12} & Y_{1m} \\ Y_{21} & Y_{22} & Y_{2m} \\ \vdots & \vdots & \vdots \\ Y_{n1} & Y_{n2} & Y_{nm} \end{bmatrix} = \begin{bmatrix} 1 & z_{11} & \dots & z_{1r} \\ 1 & z_{21} & \dots & z_{2r} \\ \vdots & \vdots & & \vdots \\ 1 & z_{n1} & \dots & z_{nr} \end{bmatrix} \begin{bmatrix} \beta_{01} & \beta_{02} & \beta_{0m} \\ \beta_{11} & \beta_{12} & \beta_{1m} \\ \vdots & \vdots & \vdots \\ \beta_{r1} & \beta_{r2} & \beta_{rm} \end{bmatrix} + \begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{1m} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{2m} \\ \vdots & \vdots & \vdots \\ \varepsilon_{n1} & \varepsilon_{n2} & \varepsilon_{nm} \end{bmatrix}$$

$\mathbf{Y} = [Y_1 \ Y_2 \ \dots \ Y_m]$ = vector of response variables (multivariate random variable)

$\mathbf{z} = [1 \ z_1 \ \dots \ z_r]$ = vector of common predictor variables (deterministic)

(usually continuous, but can also be categorical to include MANOVA-model)

$\boldsymbol{\beta} = \{\beta\}_{ij}$ = matrix of unknown regression coefficients (deterministic)

$\boldsymbol{\varepsilon} = [\varepsilon_1 \ \varepsilon_2 \ \dots \ \varepsilon_m]$ = vector of model errors (multivariate random variable)

$$\mathbf{Y} = \mathbf{Z} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Assumptions on covariance structure

Error vector elements within same observation (row) can be correlated but error vectors between different observations are uncorrelated:

$$\boldsymbol{\varepsilon} \stackrel{\text{def}}{=} \left\{ \leftarrow \boldsymbol{\varepsilon}(\mathbf{j}, :) \rightarrow \right\}_{j=1, \dots, n}$$

$$\text{Cov}[\boldsymbol{\varepsilon}(\mathbf{j}_1, :), \boldsymbol{\varepsilon}(\mathbf{j}_2, :)] = \delta_{j_1 j_2} \boldsymbol{\Sigma} = \begin{cases} \boldsymbol{\Sigma} & j_1 = j_2 \\ \mathbf{0} & j_1 \neq j_2 \end{cases}$$

$$\boldsymbol{\varepsilon} \stackrel{\text{def}}{=} \left\{ \begin{array}{c} \uparrow \\ \boldsymbol{\varepsilon}(:, \mathbf{i}) \\ \downarrow \end{array} \right\}_{i=1, \dots, m}$$

$$\text{Cov}[\boldsymbol{\varepsilon}(:, \mathbf{i}_1), \boldsymbol{\varepsilon}(:, \mathbf{i}_2)] = \sigma_{i_1 i_2} \mathbf{I}_{nn}$$

Estimates

$$\hat{\beta} = (Z^T Z)^{-1} Z^T Y \quad (\text{LS estimate})$$

$$E[\hat{\beta}] = \beta$$

$$\text{Cov}[\hat{\beta}(:, i_1), \hat{\beta}(:, i_2)] = \sigma_{i_1 i_2} (Z^T Z)^{-1}$$

$$\hat{\varepsilon} = Y - \hat{Y} = Y - Z\hat{\beta}$$

$$\hat{\Sigma} = \frac{1}{n-(r+1)} \hat{\varepsilon}^T \hat{\varepsilon}$$

Distribution assumption for inference

$$\varepsilon(j, :) \sim N_m(0, \Sigma) \quad \text{giving}$$

$$\hat{\beta}(:, i) \sim N_{r+1}(\beta(:, i), \sigma_{ii}(Z^T Z)^{-1})$$

Extra Sum of Squares (ESS) test for significant predictor variables

Split up $\beta = \begin{bmatrix} \beta_{(1)} \\ \beta_{(2)} \end{bmatrix}$ where β is $(r+1) \times m$, $\beta_{(1)}$ is $(q+1) \times m$, $\beta_{(2)}$ is $(r-q) \times m$

$H_0: \beta_{(2)} = 0$ (meaning corresponding predictor variables are insignificant)

UMLR Test statistic
$$\frac{ESS(\beta_{(1)} \rightarrow \beta) / \Delta df}{\hat{\sigma}^2} = \frac{ESS(\beta_{(1)} \rightarrow \beta) / (r-q)}{\hat{\sigma}^2} \sim F(r-q, n-(r+1))$$

MMLR version

Full model: $SSE = \hat{\varepsilon}^T \hat{\varepsilon} = (Y - Z\hat{\beta})^T (Y - Z\hat{\beta})$

H_0 model: $SSE_{H_0} = \hat{\varepsilon}_{H_0}^T \hat{\varepsilon}_{H_0} = (Y - Z_{(1)}\hat{\beta}_{(1)})^T (Y - Z_{(1)}\hat{\beta}_{(1)})$

$ESS(\beta_{(1)} \rightarrow \beta) = SSE_{H_0} - SSE = \hat{\varepsilon}_{H_0}^T \hat{\varepsilon}_{H_0} - \hat{\varepsilon}^T \hat{\varepsilon} = n\hat{\Sigma}_{(1),ML} - n\hat{\Sigma}_{ML}$

Likelihood Ratio Test statistic
$$\Lambda \stackrel{\text{def}}{=} \frac{\max_{\beta(1), \Sigma} L(\beta(1), \Sigma)}{\max_{\beta, \Sigma} L(\beta, \Sigma)} = \left(\frac{\hat{\Sigma}}{\hat{\Sigma}_{(1)}} \right)^{\frac{n}{2}}$$

$0 \leq \Lambda \leq 1$, **Reject H_0 for $\Lambda \cong 0$**

$-2 \log \Lambda = -n \log \frac{\hat{\Sigma}}{\hat{\Sigma}_{(1)}} \sim \chi^2(m(r-q))$ (large sample approximation)

Improved approximation
$$-(n - (r+1) - \frac{m-r+q+1}{2}) \log \frac{\hat{\Sigma}}{\hat{\Sigma}_{(1)}} \sim \chi^2(m(r-q))$$